

INMO(1st) – 1986

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions.
1. A person who left home between 4 p.m. and 5 p.m. returned between 5 p.m. and 6 p.m. and found that the hands of his watch had exactly exchanged place, when did he go out ?
 2. Solve

$$\log_2 x + \log_4 y + \log_4 z = 2$$

$$\log_3 x + \log_9 y + \log_9 z = 2$$

$$\log_4 x + \log_{16} y + \log_{16} z = 2$$

3. Two circles with radii a and b respectively touch each other externally. Let c be the radius of a circle that touches these two circles as well as a common tangent to the two circles. Prove that

$$\frac{1}{\sqrt{c}} = \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{b}}.$$
4. Find the least natural number whose last digit is 7 such that it becomes 5 times larger when this last digit is carried to the beginning of the number.
5. If $P(x)$ is a polynomial with integer coefficients and a, b, c , three distinct integers, then show that it is impossible to have $P(a) = b$, $P(b) = c$, $P(c) = a$.
6. Construct a quadrilateral which is not a parallelogram, in which a pair of opposite angles and a pair of opposite sides are equal.
7. If a, b, x, y are integers greater than 1 such that a and b have no common factor except 1 and $x^a = y^b$ show that $x = n^b, y = n^a$ for some integer n greater than 1.
8. Suppose $A_1, A_2, \dots, A_6$ are six sets each with four elements and $B_1, B_2, \dots, B_n$ are n sets each with two elements, Let $S = A_1 \cup A_2 \cup \dots \cup A_6 = B_1 \cup B_2 \cup \dots \cup B_n$. Given that each elements of S belongs to exactly four of the A 's and to exactly three of the B 's, find n .
9. Show that among all quadrilaterals of a given perimeter the square has the largest area.

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